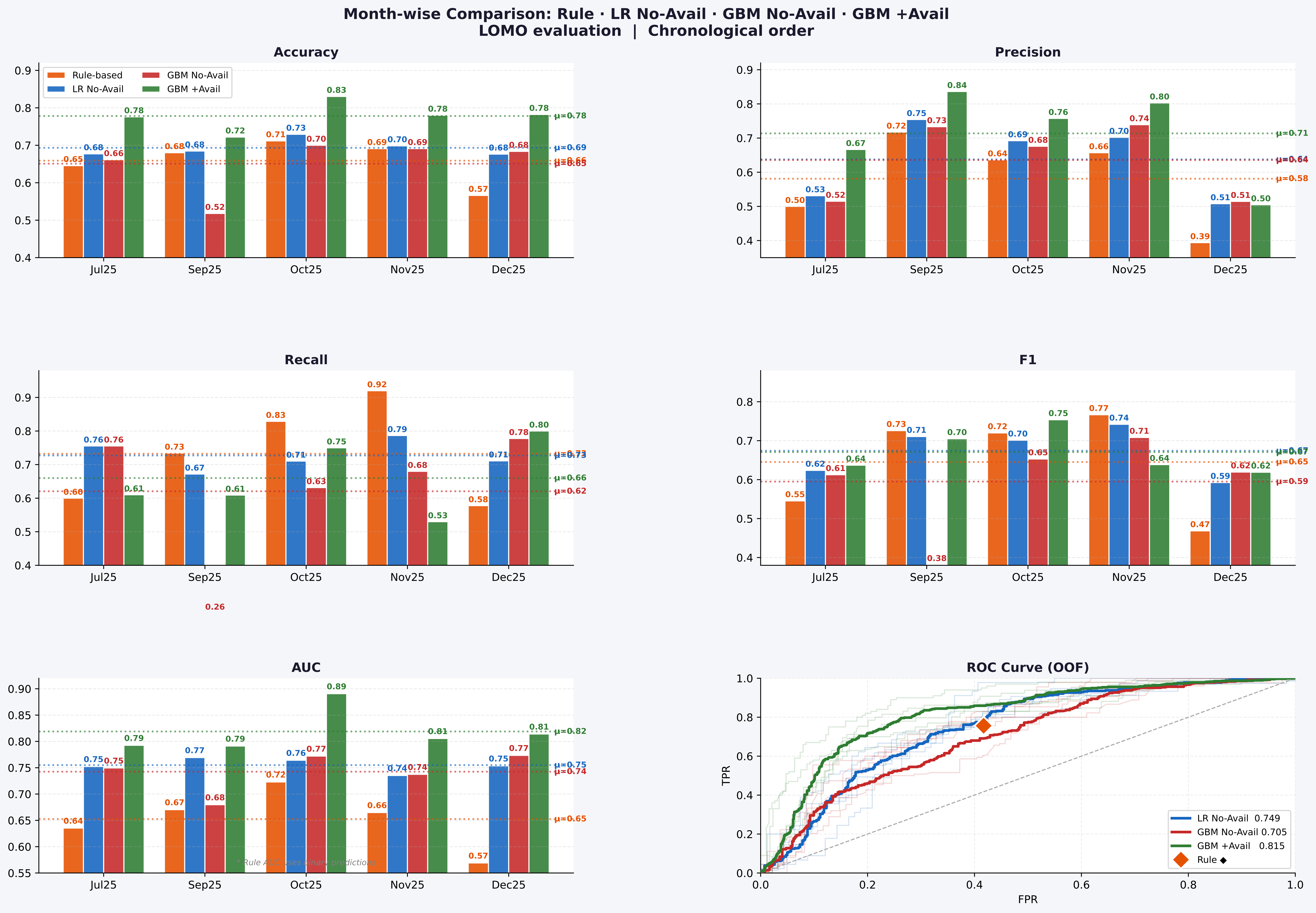
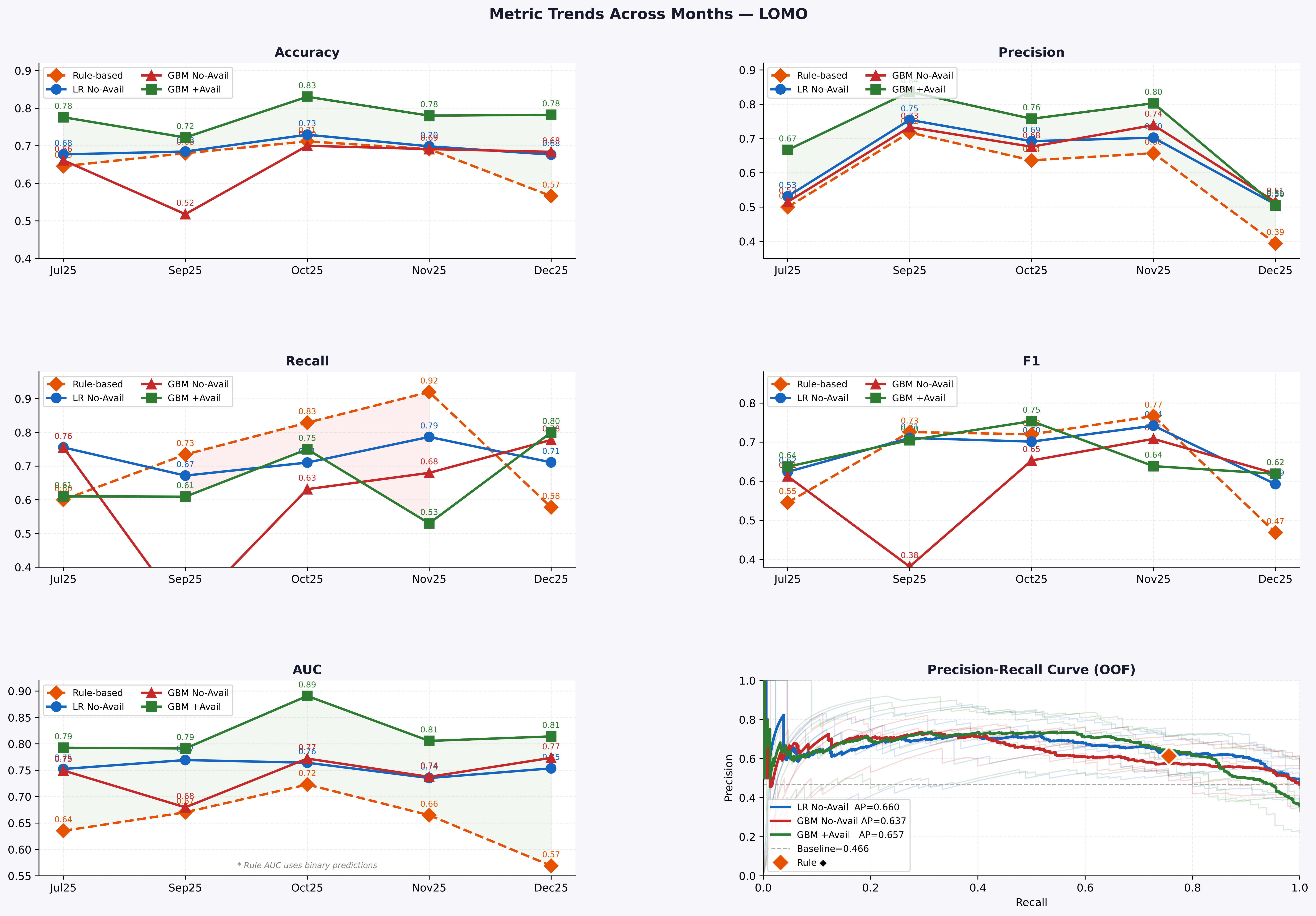
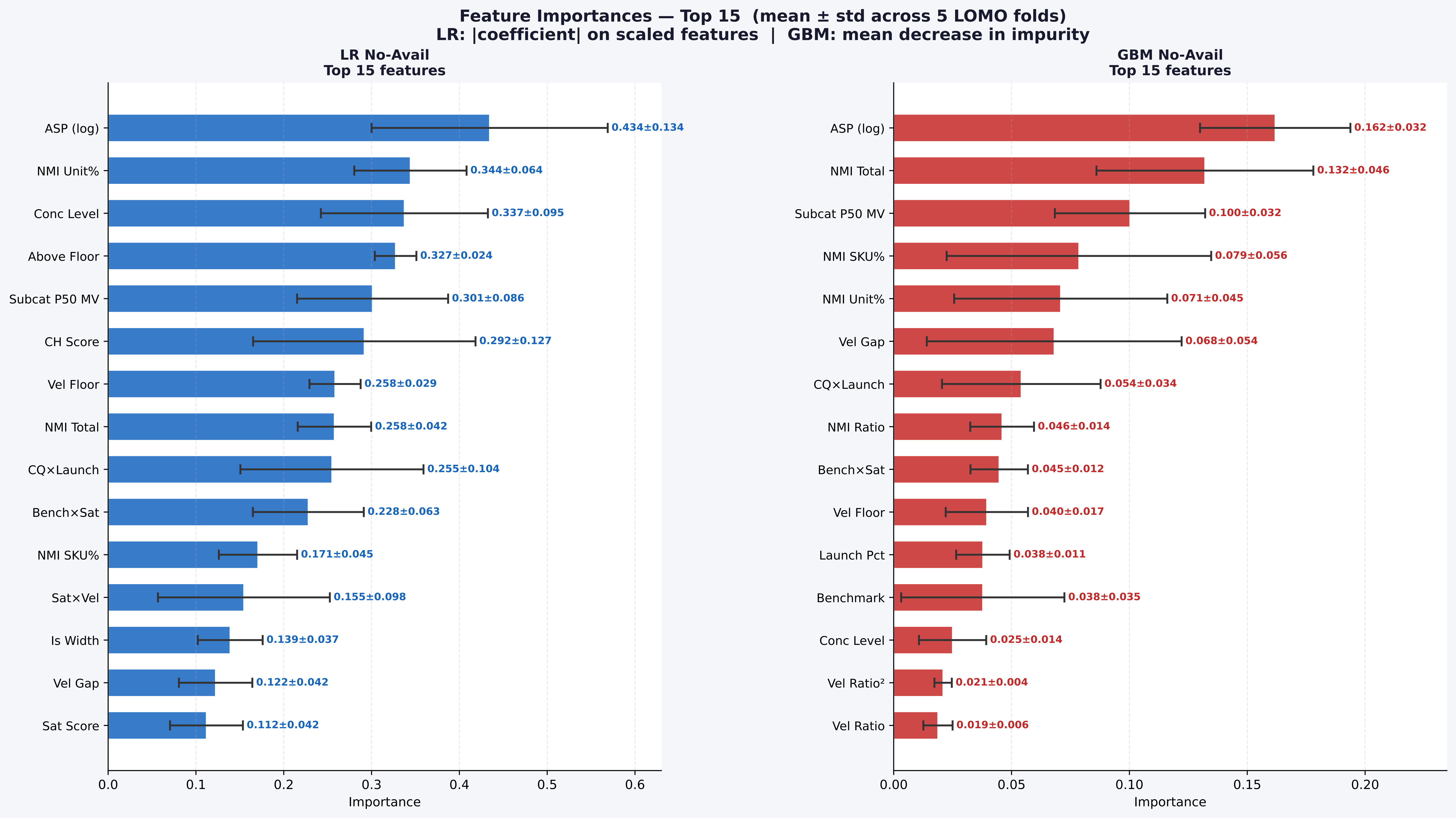

Month-wise Comparison: Rule · LR No-Avail · GBM No-Avail · GBM +Avail
LOMO evaluation | Chronological order
Accuracy
Precision
Recall
F1
AUC
ROC Curve (OOF)
Rule-based
LR No-Avail
GBM No-Avail
GBM +Avail
Jul25
Sep25
Oct25
Nov25
Dec25
* Rule AUC uses binary predictions
LR No-Avail 0.749
GBM No-Avail 0.705
GBM +Avail 0.815
Rule ◆
TPR
FPR


Metric Trends Across Months — LOMO
Accuracy
Precision
Recall
F1
AUC
Rule-based
LR No-Avail
GBM No-Avail
GBM +Avail
Jul25
Sep25
Oct25
Nov25
Dec25
* Rule AUC uses binary predictions
Precision-Recall Curve (OOF)
Precision
Recall
LR No-Avail AP=0.660
GBM No-Avail AP=0.637
GBM +Avail AP=0.657
Baseline=0.466
Rule ◆

Full Month-wise Delta Table — All Metrics | LOMO | Chronological Order

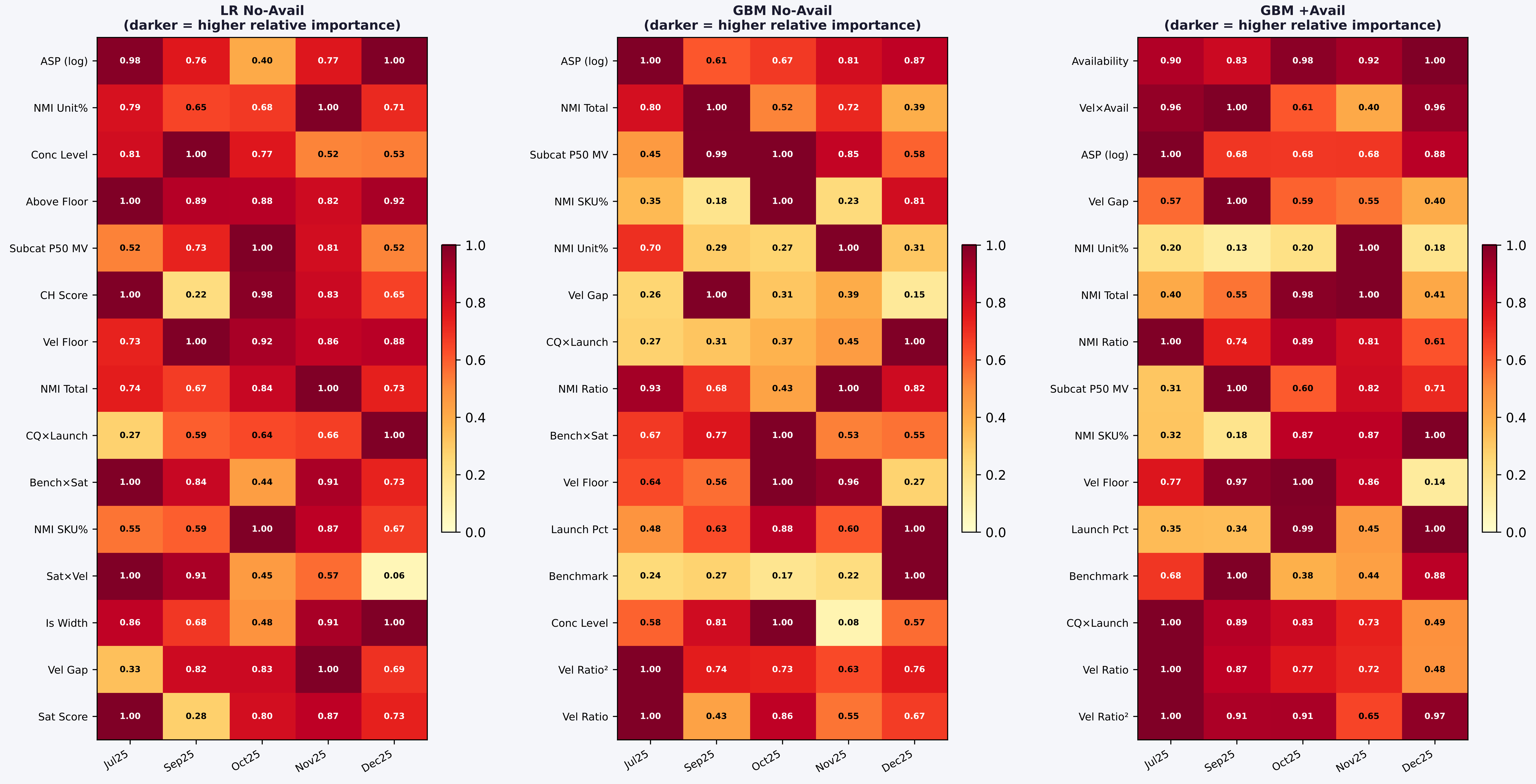
Month	Rule Acc	LR Acc	Δ	GBM Acc	Δ	GBM+Av Acc	Δ	Rule Prec	LR Prec	Δ	GBM Prec	Δ	GBM+Av Prec	Δ	Rule Rec	LR Rec	Δ	GBM Rec	Δ	GBM+Av Rec	Δ	Rule F1	LR F1	Δ	GBM F1	Δ	GBM+Av F1	Δ	LR AUC	GBM AUC	GBM+Av AUC
Jul25	0.65	0.68	**+0.03**	0.66	**+0.02**	0.78	**+0.13**	0.50	0.53	**+0.03**	0.52	**+0.02**	0.67	**+0.17**	0.60	0.76	**+0.16**	0.76	**+0.16**	0.61	**+0.01**	0.55	0.62	**+0.08**	0.61	**+0.07**	0.64	**+0.09**	0.752	0.750	0.793
Sep25	0.68	0.68	**+0.00**	0.52	**-0.16**	0.72	**+0.04**	0.72	0.75	**+0.04**	0.73	**+0.02**	0.84	**+0.12**	0.73	0.67	**-0.06**	0.26	**-0.48**	0.61	**-0.13**	0.73	0.71	**-0.02**	0.38	**-0.34**	0.70	**-0.02**	0.769	0.680	0.791
Oct25	0.71	0.73	**+0.02**	0.70	**-0.01**	0.83	**+0.12**	0.64	0.69	**+0.06**	0.68	**+0.04**	0.76	**+0.12**	0.83	0.71	**-0.12**	0.63	**-0.20**	0.75	**-0.08**	0.72	0.70	**-0.02**	0.65	**-0.07**	0.75	**+0.03**	0.764	0.772	0.891
Nov25	0.69	0.70	**+0.01**	0.69	**+0.00**	0.78	**+0.09**	0.66	0.70	**+0.05**	0.74	**+0.08**	0.80	**+0.15**	0.92	0.79	**-0.13**	0.68	**-0.24**	0.53	**-0.39**	0.77	0.74	**-0.02**	0.71	**-0.06**	0.64	**-0.13**	0.735	0.737	0.806
Dec25	0.57	0.68	**+0.11**	0.68	**+0.12**	0.78	**+0.22**	0.39	0.51	**+0.11**	0.51	**+0.12**	0.50	**+0.11**	0.58	0.71	**+0.13**	0.78	**+0.20**	0.80	**+0.22**	0.47	0.59	**+0.12**	0.62	**+0.15**	0.62	**+0.15**	0.754	0.774	0.814
Mean	**0.66**	**0.69**	**+0.03**	**0.65**	**-0.01**	**0.78**	**+0.12**	**0.58**	**0.64**	**+0.06**	**0.64**	**+0.05**	**0.71**	**+0.13**	**0.73**	**0.73**	**-0.01**	**0.62**	**-0.11**	**0.66**	**-0.07**	**0.65**	**0.67**	**+0.03**	**0.59**	**-0.05**	**0.67**	**+0.03**	**0.755**	**0.743**	**0.819**

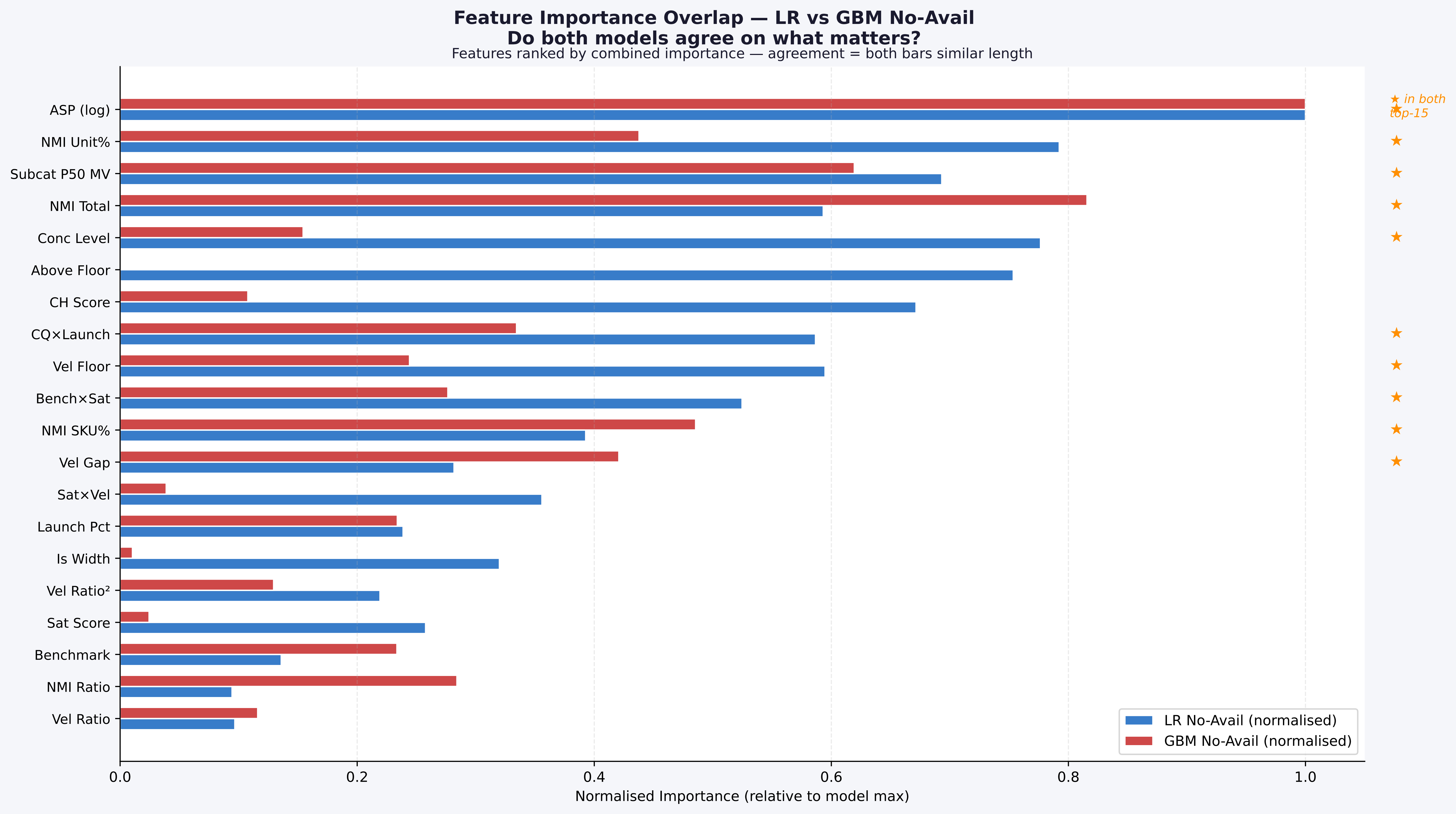
Δ = model minus Rule | green = model better | red = rule better | AUC for Rule uses binary predictions



Feature Importances — Top 15 (mean ± std across 5 LOMO folds)
LR: |coefficient| on scaled features | GBM: mean decrease in impurity
LR No-Avail
Top 15 features
ASP (log)
NMI Unit%
Conc Level
Above Floor
Subcat P50 MV
CH Score
Vel Floor
NMI Total
CQ×Launch
Bench×Sat
NMI SKU%
Sat×Vel
Is Width
Vel Gap
Sat Score
0.434±0.134
0.344±0.064
0.337±0.095
0.327±0.024
0.301±0.086
0.292±0.127
0.258±0.029
0.258±0.042
0.255±0.104
0.228±0.063
0.171±0.045
0.155±0.098
0.139±0.037
0.122±0.042
0.112±0.042
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Importance
GBM No-Avail
Top 15 features
ASP (log)
NMI Total
Subcat P50 MV
NMI SKU%
NMI Unit%
Vel Gap
CQ×Launch
NMI Ratio
Bench×Sat
Vel Floor
Launch Pct
Benchmark
Conc Level
Vel Ratio²
Vel Ratio
0.162±0.032
0.132±0.046
0.100±0.032
0.079±0.056
0.071±0.045
0.068±0.054
0.054±0.034
0.046±0.014
0.045±0.012
0.040±0.017
0.038±0.011
0.038±0.035
0.025±0.014
0.021±0.004
0.019±0.006
0.00
0.05
0.10
0.15
0.20
Importance

Feature Importance Stability Across Months — LOMO Folds
Each cell = normalised importance (row-normalised). Consistent colour = stable feature.




Feature Importance Overlap — LR vs GBM No-Avail
Do both models agree on what matters?
Features ranked by combined importance — agreement = both bars similar length
ASP (log)
NMI Unit%
Subcat P50 MV
NMI Total
Conc Level
Above Floor
CH Score
CQ×Launch
Vel Floor
Bench×Sat
NMI SKU%
Vel Gap
Sat×Vel
Launch Pct
Is Width
Vel Ratio²
Sat Score
Benchmark
NMI Ratio
Vel Ratio
★ in both top-15
LR No-Avail (normalised)
GBM No-Avail (normalised)
0.0
0.2
0.4
0.6
0.8
1.0
Normalised Importance (relative to model max)